OPERATING SYSTEMS / MOCK EXAM 05

80 minutes | 100 points | Name: _____

Show your work. Use the exact model and event rules specified in each question.

1. Controlled kernel entry (12 points)

(a) A user program loops forever without system calls. Trace how the OS can regain control. (4 points)

(b) Can a trap return to the same process without a process context switch? Give a concrete execution. (4 points)

(c) Distinguish interrupt-vector selection from system-call-number dispatch. (4 points)

2. A race despite coherent caches (12 points)

Two cores share x=10. Each thread executes one x=x+1 as separate load, add, store operations. Memory is sequentially consistent and caches are coherent.

(a) Give a legal interleaving that loses an update and its final x. (5 points)

(b) State exactly what must be atomic or protected. (3 points)

(c) Explain why cache coherence alone does not prevent this race. (4 points)

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3. STCF with staggered arrivals (14 points)

One CPU, no I/O or switching cost. Jobs (arrival, CPU time): A(0,7), B(2,3), C(3,1), D(6,2). Ties use earlier arrival; preempt only for strictly less remaining time.

(a) Give the complete STCF timeline. (6 points)

(b) Give completion times and turnaround times for all jobs. (4 points)

(c) Give mean turnaround and mean response. (4 points)

4. Sleeping lock handoff (12 points)

A waiting thread appends itself to a queue under guard, releases guard, and calls park(). An early unpark() is lost unless setpark() has been called.

(a) Construct a lost-wakeup execution with an owner and a waiter. (4 points)

(b) Place setpark() precisely and state its required effect. (4 points)

(c) During direct handoff, why must flag remain 1? Does guard itself disable preemption? (4 points)

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5. Semaphore signals and readers (12 points)

(a) done starts at 0. Three workers each post(done) once after writing their private result. The parent performs three waits before reading all results. Is this safe if all posts occur first? Explain. (4 points)

(b) In a reader-preference lock, readers increment/decrement a shared count; the first reader acquires write_lock and the last releases it. What must the count lock protect? (4 points)

(c) Give an arrival pattern that can starve a writer. (4 points)

6. Deadlock, livelock, and ordering (12 points)

Three single-instance locks cannot be forcibly reclaimed. T1 holds L1 and waits for L2; T2 holds L2 and waits for L3; T3 holds L3 and waits for L1.

(a) Map all four deadlock conditions to this state. (4 points)

(b) Give one global acquisition order that prevents this cycle, and identify the removed condition. (4 points)

(c) A retry scheme releases all held locks after a failed trylock and immediately retries. What progress failure remains, and what can reduce synchronized retries? (4 points)

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7. Parallel performance budget (12 points)

A fixed job takes 240 s on one core; 25% is serial. Parallel work divides perfectly. On N>1 cores, extra elapsed overhead is 2(N-1) s; no overhead on one core.

(a) Find elapsed time and speedup for N=4 and N=8. (6 points)

(b) Find the infinite-core speedup bound if overhead is ignored. (3 points)

(c) With the stated overhead, which of N=4 and N=8 is faster? Justify numerically. (3 points)

8. Queues, migration, and locality (14 points)

Two cores use private FIFO queues. CPU0 has A(10), B(4); CPU1 has C(3), D(2). All arrive at 0. Jobs are nonpreemptive. CPU1 may steal B only after its queue becomes empty. A steal costs CPU1 1 unit before B runs; CPU0 keeps running. Other overhead is zero.

(a) Give both timelines and makespan without stealing and with stealing. (8 points)

(b) Explain one locality cost of migration and one shared-queue scalability cost. (4 points)

(c) Does having per-core queues make every stealing operation automatically lock-free? Explain. (2 points)